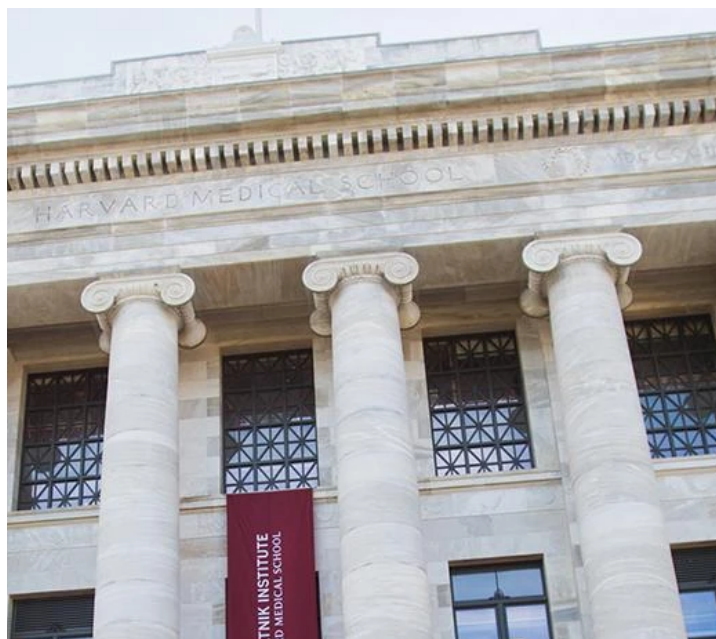


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October 2024



Providing Thoughtful Leadership on AI in Medicine



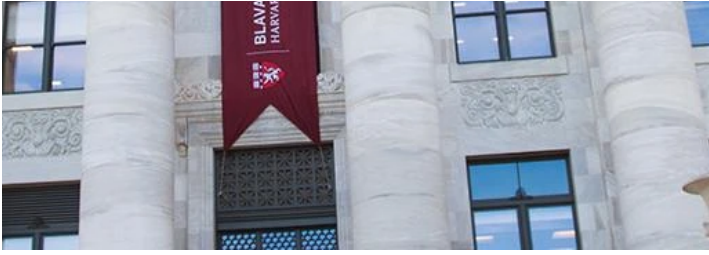
A commentary

Fall 2024

 by George Q. Daley

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In 1996, this magazine devoted an issue to an emerging technology that seemed poised to reshape science, medicine, and even society at large.

"The World Wide Web may seem a curiosity to some, an object of hype to many, even a danger to others," wrote Robert Greenes, MD '66 PhD '70. Jerome Kassirer, then the editor in chief of the *New England Journal of Medicine*, wrote, "Although a health care delivery system that depends, even partly, on online communications holds considerable promise, the problems it poses are enormous.... For the benefit of our patients, physicians should be at the forefront of these changes, not dragged along by progress."

It is not hard to see the parallels to today's debates over generative artificial intelligence, perhaps the technology with the greatest potential to change how science and medicine are conducted since the internet. As with the internet, the use of generative AI in science and medicine has both tremendous

potential and serious risks. HMS is already leading the way in applying a host of AI technologies creatively and thoughtfully, and we are well positioned to do the same with generative AI. Implemented responsibly, tools powered by generative AI can improve the care of patients, speed the development of therapeutics, and help us gain a deeper understanding of scientific mysteries. The key is to ensure that human experts – physicians and scientists – are at the center of these efforts.

I am impressed by the work being done by faculty in our Department of Biomedical Informatics and clinical faculty at our affiliated hospitals to understand the strengths and limitations of AI in medicine. They are providing invaluable insights into how to empower physicians and proving that, although it is a relatively new technology, generative AI can be evaluated using long-standing techniques, as with any other intervention.

I am also encouraged by the response we received to our call for proposals for funding to investigate the use of generative AI in education, research, and administration. Earlier this year, we awarded funding to thirty-three teams of faculty, clinicians, students, and staff. Many of these projects focus on generative AI in medical education, including developing chatbots to simulate patient interactions and provide personalized feedback to our students.

Understanding and evaluating information and consultation powered by large language models is going to become a vital skill for the physicians of tomorrow. To that end, we have built AI into the medical curriculum and will continue to adapt it. We've designed a course on AI in medicine for incoming students in our Health Sciences and Technology program, and students in our Pathways track are using large language models as tutors. This fall also marks the launch of a new PhD track in AI in medicine that recently welcomed its first class to campus.

As Greenes and Kassirer predicted almost thirty years ago, the internet is now ubiquitous in science and medicine. I believe the same will soon be true of generative AI. The leadership of today's HMS faculty and students is crucial to ensuring that happens safely and effectively.

George Q. Daley is the dean of Harvard Medical School.

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